



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

09-05-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE/ALL STAR BATCH/(SCORE ADVANCED)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C,D	A,B,D	C,D	A,B	A,B,C,D	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	D	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.16 to 0.19	0.11 to 0.13	0.4	0.5	2.4	6.05 to 6.06	4.4 to 4.6	0.27 to 0.29

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	A,B,C,D	A,C	A,B	A,B,C,D	D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.00	3.00	10.00	138.00	4.00	1.00	348.16	40.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,B,C	A,C,D	A,B	A,B,D	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	C	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.00	13.00	7.00	1.00	8.00	3.00	8.00	4.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B,C,D)

If they collide, their vertical component of velocities should be same,

$$\text{i.e., } 100\sin\theta = 160\sin 30^\circ$$

$$\Rightarrow \sin\theta = 4/5$$

Their vertical components will always be same.
Horizontal components.

$$\text{And } 160\cos 30^\circ = 80\sqrt{3} \text{ ms}^{-1}$$

They are not same, hence their velocities will not be same at any time. So (b) is correct.

$$x = x_1 - x_2 = 160\cos 30^\circ t - 100\cos\theta t$$

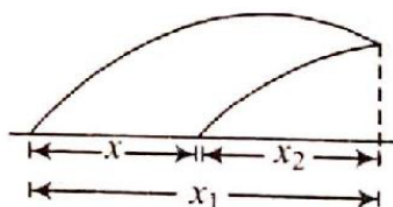
$$\text{Time of flight: } T = \frac{2 \times 160 \times \sin 30^\circ}{g}$$

$$= 16\text{s (same for both)}$$

Now $t < T_x \rightarrow$ to collide in air

$$\Rightarrow \frac{x}{80\sqrt{3} - 60} < 16 \Rightarrow x < 1280\sqrt{3} - 960$$

Since their times of flight are same, they will simultaneously reach their maximum height. So it is possible to collide at the highest point for certain values of x .



2. Ans (A,B,D)

According to Pascal's law, $P_D = P_B$

Because B and D lie on the same horizontal plane.

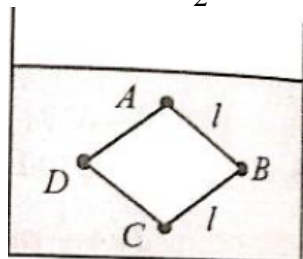
Let ℓ be the side of the square. Then the vertical

distance between A and C is $\frac{\ell}{\sqrt{2}}$

$$P_B = P_V = P_C - \rho g \frac{L}{\sqrt{2}} \dots\dots (i)$$

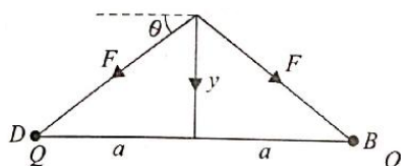
$$\text{Similarly, } P_B = P_D - P_A + \rho g \frac{L}{\sqrt{2}} \dots\dots\dots (ii)$$

$$P_B = P_D = \frac{P_A + P_C}{2}$$



3. Ans (C,D)

The particle will oscillate or perform SHM if equilibrium is a stable one. For negative charge, equilibrium is stable, if the particles are displaced along CD and unstable for displacement along AB.



For $y \ll a$, in displaced position, net resultant force toward equilibrium position is

$$F_{\text{res}} = 2F \sin \theta = \frac{2 \times Q}{4\pi\epsilon_0(a^2 + y^2)} \times \frac{y}{(a^2 + y^2)^{1/2}} = \frac{2Qy}{4\pi\epsilon_0 a^3} [a \gg y]$$

$$F_{\text{res}} \propto y, \text{ so SHM for } y \ll a$$

For $y = a$, there would be restoring force, but F is not proportional to y . So there is oscillating motion but not SHM.

4. Ans (A,B)

Let d = distance of the target T from the point of projection. P will strike T if d is an integral multiple of the pitch.

$$\text{Pitch} = \left(\frac{2\pi m}{QB} \right) v \cos \theta = k \left(\frac{v}{B} \right)$$

Where k = constant

$$\text{Initially, pitch } d = k \left(\frac{v_0}{B_0} \right)$$

For the given options if pitch $d' = \frac{d}{2}, \frac{d}{3}, \frac{d}{4}, \dots$ etc, then change will hit the target

5. Ans (A,B,C,D)

Δx at $O = d$ [path difference is maximum at O]

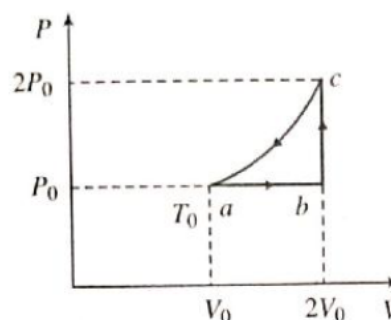
So, if $d = \frac{7\lambda}{2}$, O will be a minima and hence intensity is minimum.

If $d = 4.8\lambda$, then total 10 minima can be observed on the screen, 5 above O and 5 below O , which correspond to

$$\Delta x = \pm \frac{\lambda}{2}, \pm \frac{3\lambda}{2}, \pm \frac{5\lambda}{2}, \pm \frac{7\lambda}{2}, \pm \frac{9\lambda}{2}$$

6. Ans (A,B,C,D)

One mole of an ideal monoatomic gas (initial temperature T_0) is made to go through the cycle $abca$ shown in figure. U denotes the internal energy.



$$\text{For the process ab, } \frac{P_0 V_0}{T_0} = \frac{P_0/2 \cdot 2V_0}{T_b}$$

$$T_b = 2T_0$$

$$T_b > T_a \Rightarrow U_b > U_a$$

$$U_b - U_a = C_V \Delta T = \frac{3R}{2} (2T_0 - T_0) = \frac{3RT_0}{2}$$

For the process bc,

$$\frac{P_0(2V_0)}{2T_0} = \frac{2P_0(V_0)}{T_c} \Rightarrow T_c = 4T_0$$

$$T_c > T_0$$

$$U_c - U_b = \frac{3R}{2} (4T_0 - 2T_0) = \frac{9RT_0}{2}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A)

Let the acceleration of various blocks are as shown in Figure. Pulley P_2 will have downward acceleration a .

$$\text{Now } a = \frac{a_1 + a_2}{2} \Rightarrow a_2 = 2a - a_1 > 0$$

So acceleration of 2 is upwards

Hence, (i) $\rightarrow$ (b, c)

$$\text{And } a = \frac{-a_1 + a_4}{2} \Rightarrow a_4 = 2a + a_1 > 0$$

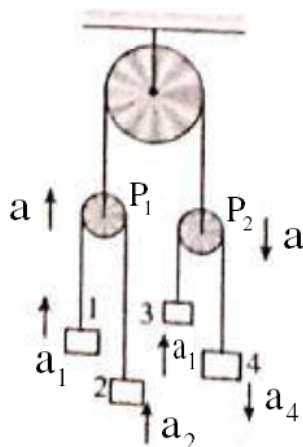
So, acceleration of 4 is downwards.

Hence, (ii) $\rightarrow$ (a, d)

Acceleration of 2 w.r.t 4

$$a_{\frac{2}{4}} = a_2 - (-a_4) = 4a > 0$$

This is upwards, Hence, (iv) $\rightarrow$ (c)



9. Ans (D)

$$(P) n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3 \\ = 10 : 80 : 10 = 0.5 : 4 : 0.5$$

$$\Rightarrow \frac{5\lambda}{2} = 1 \Rightarrow \lambda = 0.4\text{m}$$

$$V = \sqrt{\frac{Y}{\rho}} = \frac{5000}{0.4} = 12500 = 12.5 \text{ kHz}$$

$$(Q) n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3 \\ = 30 : 40 : 30 = 1.5 : 2 : 1.5$$

$$\Rightarrow \frac{5\lambda}{2} = 1 \Rightarrow \lambda = 0.4\text{m}$$

$$V = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{1.6 \times 10^{11}}{2500}} = 8000\text{m/s}$$

$$f = \frac{v}{\lambda} = \frac{8000}{0.4} = 20000 = 20 \text{ kHz}$$

$$(R) n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3 \\ = 10 : 60 : 30 = 0.5 : 3 : 1.5$$

$$\Rightarrow \frac{5\lambda}{2} = 1 \Rightarrow \lambda = 0.4\text{m}$$

$$V = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{2 \times 10^{11}}{8000}} = 5000\text{m/s}$$

$$f = \frac{v}{\lambda} = \frac{5000}{0.4} = 12500 = 12.5 \text{ kHz}$$

$$(S) n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3 \\ = 15 : 60 : 25 = 1.5 : 6 : 2.5$$

$$\Rightarrow \frac{10\lambda}{2} = 1 \Rightarrow \ell = 0.2\text{m}$$

$$V = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{1.6 \times 10^{11}}{2500}} = 8000 \text{ m/s}$$

$$f = \frac{v}{\lambda} = \frac{8000}{0.2} = 40000 = 40 \text{ kHz}$$

10. Ans (D)

Maximum kinetic energy of ejected electron is given by Einstein's photoelectric equation

$$K_{\max} = hf - \left(\frac{hf}{3} \right) = \frac{2hf}{3}. \text{ As no potential difference is applied across target and collector and vacuum is there in the tube, so this maximum KE remains same at all locations. b, c and d. Kinetic energy of ejected photoelectrons can be anything from 0 to } K_{\max} \text{ (as found for a). It remains the same at all locations (reasoning is same as for above).}$$

PART-1 : PHYSICS

SECTION-II

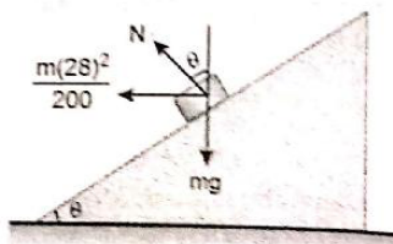
1. Ans (0.16 to 0.19)

When there is not tendency to slip, no frictional force acts. Hence

$$N \sin \theta = \frac{m(28)^2}{200} \dots\dots 1$$

$$\text{and } N \cos \theta = mg \dots\dots 2$$

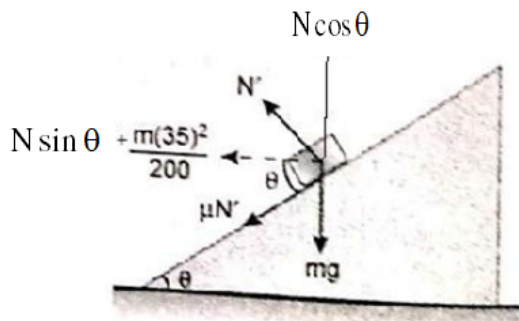
Dividing (1) by (2), we get $\tan \theta = 0.4 \dots\dots 3$



At speed 35 m/s, car has a tendency to slip up the track. Hence frictional force acts down the plane.

$$\text{Therefore, } N' \sin \theta + \mu N' \cos \theta = \frac{m(35)^2}{200} \dots\dots 4$$

$$\text{and } N' \cos \theta - \mu N' \sin \theta = mg \dots\dots 5$$



Solving (3), (4) and (5), we get $\mu = 0.18$

2. Ans (0.11 to 0.13)

$$\text{Initial kinetic energy } \frac{1}{2} \times 2 \times 10^2 = 100 \text{ J}$$

$$\text{Final KE} = 50 \text{ J}$$

$$\therefore \frac{1}{2} m v^2 = 50 \text{ J}$$

When the spring is being compressed, the equation of motion of spring mass system can be written as

$$x A \sin \omega t \text{ where } \omega = \sqrt{\frac{k}{m}}$$

$$\therefore \frac{dx}{dt} = A \omega \cos \omega t$$

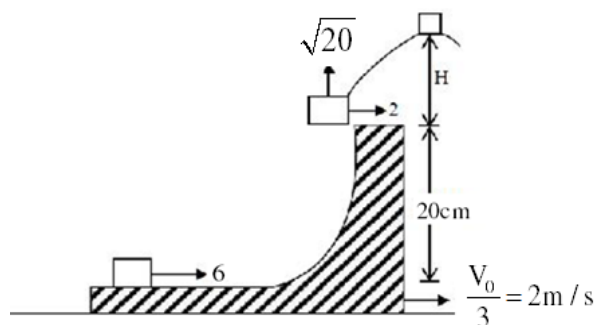
$$\Rightarrow v = v_0 \cos \omega t$$

$$\Rightarrow 5\sqrt{2} = 10 \cos \omega t$$

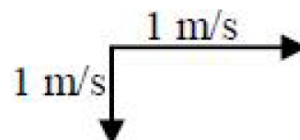
$$\Rightarrow \omega t = \frac{\pi}{4}$$

$$\Rightarrow t = \frac{\pi}{4\omega} = \frac{\pi}{4} \sqrt{\frac{m}{k}} = \frac{\pi}{4} \sqrt{\frac{2}{80}} = 0.124 \text{ sec}$$

3. Ans (0.4)



At highest point combined mass velocity is



$$H = \frac{20}{2g} = 1 \text{ m}$$

$$1.2 = 1.T + \frac{1}{2} g T^2$$

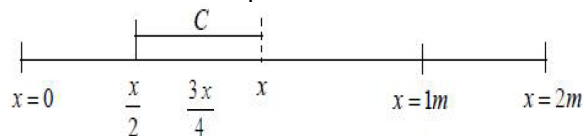
$$T = 0.4 \text{ sec}$$

$$u_{\text{rel}} = 1 \text{ m/s}$$

$$\Rightarrow \ell = u_{\text{rel}} T = 0.4 = 40 \text{ cm}$$

4. **Ans (0.5)**

The co-ordinates of centre of mass (point C) of the moving part is $\frac{3x}{4}$



$$\therefore X_c = \frac{3x}{4}$$

$\therefore$

$$v_c = \frac{d}{dt}(X_c) = \frac{3}{4} \frac{dx}{dt} = \frac{3}{4} \text{ m/s} \left(\text{as } \frac{dx}{dt} = 1 \text{ m/s} \right)$$

Linear momentum of moving point is $P = mv$

Where $v = 1 \text{ m/s}$ and m is increasing uniformly with time.

The net force on the moving part is thus,

$$F = \frac{dP}{dt} = \frac{dm}{dt}v + \frac{dv}{dt}m$$

$$\text{or } F = \left(\frac{dm}{dt} \right) (1) + 0$$

Here $\frac{dm}{dt}$ can be found with the help of following argument.

The moving end of the carpet starts from the origin and the whole carpet will be moving when it reaches $x = 2 \text{ m}$. This will happen after 2 seconds. Hence $\frac{dm}{dt} = \frac{1}{2} \text{ kg/s}$

$$\therefore \frac{dm}{dt} = \frac{1}{2} \text{ kg/s}$$

$$\therefore F_{\min} (\text{neglecting all dissipative forces}) = \frac{1}{2} \text{ N}$$

5. **Ans (2.4)**

Bulk modulus is given by $B = -\frac{dP}{(dV/V)}$ or

$$\frac{dV}{V} = -\frac{dP}{B}$$

$$\text{or } P \cdot \frac{dV}{V} = -\frac{PdP}{B}$$

$\frac{PdV}{V}$ is the work done per unit volume. The negative sign implies that a decrease in pressure gives rise to increase in volume and vice-versa.

Hence volume density of elastic potential

$$\text{energy } u = \int_0^{hpg} \frac{P \cdot dP}{B} = \frac{1}{2} \frac{(hpg)^2}{B}$$

Substitution the value, we obtain

$$u = \frac{1}{2} \frac{(10^3)(10^3)^2 \cdot (9.8)^2}{2 \times 10^9}$$

$$\Rightarrow u = 2.4 \times 10^4 \text{ J/m}^3$$

6. **Ans (6.05 to 6.06)**

Applying Snell's law on face AB,

$$1 \sin 45^\circ = \sqrt{2} \sin r$$

$$\therefore \sin r = \frac{1}{2}$$

$$\text{or } r = 30^\circ$$

i.e., ray becomes parallel to AD inside the block.

Now applying, $\frac{\mu_0}{v} = \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ on face

$$CD = \frac{1.514}{OE} - \frac{\sqrt{2}}{\infty} = \frac{1.514 - 1.414}{0.4}$$

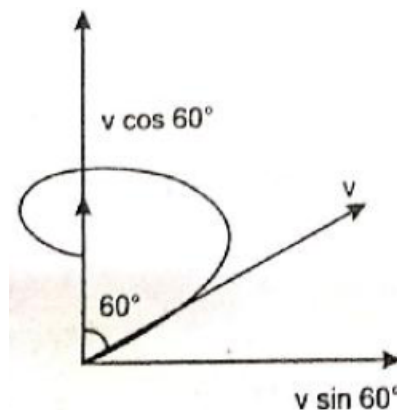
Solving this equation, we get $OE = 6.06 \text{ m}$

7. **Ans (4.4 to 4.6)**

Pitch of the helix is given by

$$y = (v \cos \theta) \cdot T = v(\cos 60^\circ)T = \frac{vT}{2}$$

Where time period of circular motion $T = \frac{2\pi m}{eB}$



$$\text{Pitch } y = \left(\frac{2\pi m}{Be} \right) v \cos 60^\circ = \left(\frac{\pi m v}{Be} \right)$$

$$\therefore KE = \frac{1}{2} m v^2 = \frac{1}{2} m \left(\frac{Be y}{\pi m} \right)^2$$

Using Einstein's equation

$$\frac{1}{2} m v^2 = h\nu - \phi$$

$$\phi = h\nu - \frac{1}{2} m \left(\frac{Be y}{\pi m} \right)^2$$

$$= \left[4.9 - \frac{1}{2} \left(\frac{m}{e} \right) \left(\frac{Be y}{\pi m} \right)^2 \right]$$

$$\text{electron - volts} = 4.9 - 0.4 = 4.5 \text{ eV}$$

8. **Ans (0.27 to 0.29)**

The product nucleus ^{198}Hg is in excited state and possesses extra 1.088 MeV energy.

If ^{198}Hg would had been in ground state, the kinetic energy available to electron and antineutrino must have

$$\begin{aligned}
 Q &= (m_{\text{Au}} - m_{\text{Hg}}) 931 \text{ MeV} \\
 &= (197.968233 - 197.966760) 931 \text{ MeV} \\
 &= 1.3714 \text{ MeV}
 \end{aligned}$$

Since ^{198}Hg is in excited state, actual kinetic energy available to electron and antineutrino is

$$\begin{aligned}
 K &= (1.3714 - 1.088) \text{ MeV} \\
 &= 0.283 \text{ MeV}
 \end{aligned}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (A,B,C)**

Metals of 2nd & 3rd transition series form low spin complexes with weak or strongfield ligand and form square planar complexes with complex number 4.

Ni(II) with strong ligand like CN^- form square planar complexes.

Hg(II) form tetrahedral complexes since it has d^{10} config.

2. **Ans (A,B,C,D)**

Froth floatation method is a physical method depending on adsorption of potassium ethyl xanthate which prevents the wettability of or particle. It is used only for the concentration of sulphide ores where ore particles come into frothwhile impurities sink down.

3. **Ans (A,C)**

Both $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ are not possible on vinyl halides and aryl halides.

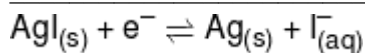
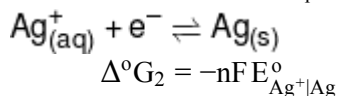
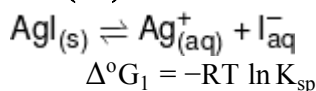
4. **Ans (A,B)**

Value based.

5. **Ans (A,B,C,D)**

Variation of composition changes internal energy of the system.

6. **Ans (D)**



$$\Delta^\circ G_3 = -nF E^\circ_{\text{I}^-/\text{AgI}|\text{Ag}}$$

$$\Delta^\circ G_3 = \Delta^\circ G_1 + \Delta^\circ G_2$$

Substituting, we get

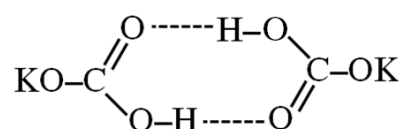
$$\log K_{\text{sp}} = \frac{-0.952}{0.059} = -16.13$$

PART-2 : CHEMISTRY

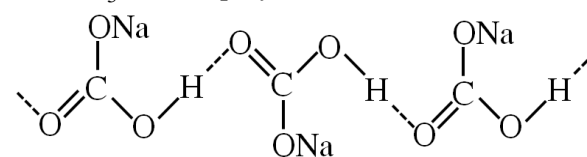
SECTION-I (ii)

7. **Ans (A)**

KHCO_3 exist as dimer.



NaHCO_3 exist in polymeric form.



LiHCO_3 exist only solution and exist as Li^+ and HCO_3^- similar to $\text{Mg}(\text{HCO}_3)_2$ (diagonal relationship).

8. **Ans (D)**

(A) $\text{H}_2\text{P}_2\text{O}_6^{2-}$ have 2 ionisable hydrogen. The phosphorous atoms are involved in sp^3 hybridization. It is non-planar species.

(B) Similarly $\text{H}_2\text{P}_2\text{O}_7^{2-}$ also have 2 ionisable hydrogens, sp^3 hybrid p atoms and non-planar structure.

(C) $\text{H}_2\text{S}_2\text{O}_5$ have 2 ionisable hydrogens, sp^3 hybrid sulphur atoms with non-planar structure.

(D) HSO_5^- ion has only one ionizable hydrogen in which S-atom is involved in sp^3 hybridization with non-planar structure. It can act as oxidizing agent.

9. **Ans (A)**

Direct reactions.

10. **Ans (B)**

All are fcc lattices. Symmetry is followed.

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (1.00)**

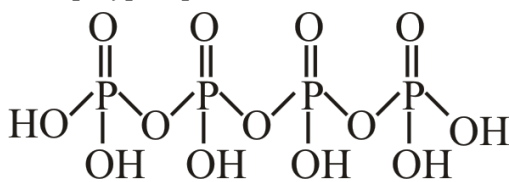
NaOH, KOH, Ba(OH)₂, Sr(OH)₂ and TlOH are strong bases. Be(OH)₂, Al(OH)₃, Sn(OH)₂ and Zn(OH)₂ are amphoteric and these react with each other forming salts that are soluble.

Mg(OH)₂ is insoluble in bases. So only Mg(OH)₂ remain in the residue. One base is excess and that also dissolve in water.

2. **Ans (3.00)**

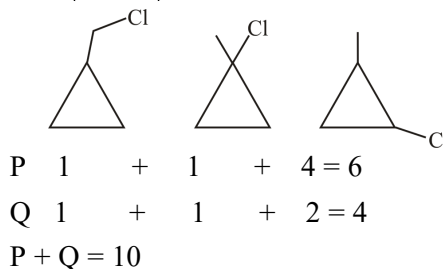
The ratio of the two different types of OH groups should be determined. If one phosphorous atom is attached by two OH groups, their K_{a1} and K_{a2} values differ.

The first OH group is a strongly acidic while the second one is weak. If only one OH group is present on one phosphorous atom. It also behave strongly acidic. So phosphorous atoms which contain two OH groups will ionize in two steps. The strongly acidic groups are titrated by the first 30.4. The two terminal OH groups are titrated by the remaining 45.6 – 30.4 = 15.2 CC. This indicates each OH requires 7.6 CC of the titrant. It can be now calculated that there are titrant. It can be now calculated that there are $\frac{30.4}{7.6} = 4$ strongly acidic groups per one polyphosphoric acid. A molecule with two terminal OH groups and four further OH groups is a tetrapolyphosphoric acid.

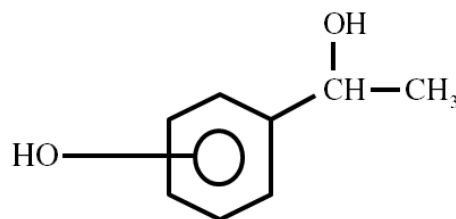


Thus the number of P – O – P bonds is 3.

3. **Ans (10.00)**



4. **Ans (138.00)**

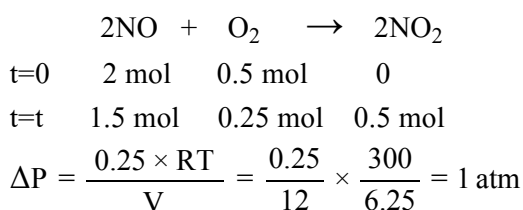


∴ minimum molecular weight = 138

5. **Ans (4.00)**

I, IV, VI, VII

6. **Ans (1.00)**



7. **Ans (348.16)**

$$IE_4(\text{Be}) = \frac{Z^2}{n^2} \times 13.6 \times 1.6 \times 10^{-19} \text{ J} = 348.16 \times 10^{-19}$$

8. **Ans (40.00)**

$$\text{Average rate in first 10 min.} = \frac{\Delta p}{\Delta t} = \frac{2}{10} \frac{\text{atm}}{\text{min}}$$

$$\text{Average rate in next 10 min.} = \frac{\Delta p}{\Delta t} = \frac{2}{10} \frac{\text{atm}}{\text{min}}$$

∴ Reaction is zero order

$$\therefore t = \frac{8}{K} = \frac{8}{1/5} = 40 \text{ mins}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C,D)

$$\int_1^{xy} f(t)dt = y \int_1^x f(t)dt + x \int_1^y f(t)dt, f(1) = 1$$

Differentiating partially w.r.t. 'y'

$$f(xy).x = \int_1^x f(t)dt + xf(y), \text{ put } y = 1$$

$$\Rightarrow x f(x) = \int_1^x f(t)dt + x$$

$$\Rightarrow f'(x) = \frac{1}{x}$$

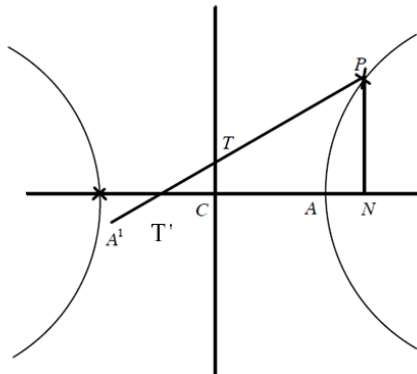
$$\Rightarrow f(x) = \begin{cases} \ln(-x) + c, & x < 0 \\ \ln x + 1, & x > 0 \end{cases}$$

$\therefore$ A, B, C, D are true

2. Ans (A,B,C)

$$C = (0, 0) \quad N = (a \sec \theta, 0)$$

Clearly



$$T = \left(0, -\frac{b}{\tan \theta}\right)$$

$$T' = \left(\frac{a}{\sec \theta}, 0\right)$$

$$CT.PN = b^2$$

$$CN.CT^1 = a^2$$

$$CN.NT^1 = AN.A^1N = a^2 \tan^2 \theta$$

$\therefore$ A, B, C are correct

3. Ans (A,C,D)

a, b, c are integers and $a^2 + b^2 + c^2 = 2 \Rightarrow$ one of them is 0 and the other can be ± 1 .

$\therefore$ No. of cases = 12

$$\text{For unique solution } |A| \neq 0 \Rightarrow \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \neq 0$$

$\therefore$ a = 0, b = 1, c = -1 and there are 6 cases

$$\therefore \text{Probability} = \frac{6}{12} = \frac{1}{2}$$

But in all the remaining 6 cases it cannot have infinitely many solutions

For example for a = 0, b = -1, c = 1, $1 \Delta_1 \neq 0$

$\therefore$ It has no solution

$\therefore$ B is not correct

a = 0 $\Rightarrow b^2 + c^2 = 2$ (1, -1) (-1, 1) it has no solution

$$a^2 + b^2 + c^2 - ab - bc - ca \geq 0$$

$$\Rightarrow ab + bc + ca \leq 2$$

$$(a + b + c) \geq 0 \Rightarrow ab + bc + ca \geq -1$$

But a = b = c not possible

$\therefore$ 2 is not there

$$ab + bc + ca \neq 0$$

$\therefore$ -1, 1 are possible values

4. Ans (A,B)

The plane after rotation is

$$\begin{vmatrix} x-3 & y-1 & z-2 \\ 2 & 4 & 4 \\ 1 & 4 & 4 \end{vmatrix} = 0$$

$$\Rightarrow (x-3)(12) - (y-1)(6) + (z-2)(0) = 0$$

$$\Rightarrow 2x - 6 - y + 1 = 0$$

$$\Rightarrow 2x - y - 5 = 0$$

Equation of plane before rotation

$$\Rightarrow \begin{vmatrix} x-3 & y-1 & z-2 \\ 1 & 2 & 2 \\ 2 & -1 & 0 \end{vmatrix} = 0$$

$$\Rightarrow 2x + 4y - 5z = 0$$

5. Ans (A, B, D)

From graph clearly it will have a solution when

$a > 0$ and when $a < 0$ then

$$\Rightarrow x^2 - a = x \text{ for atleast one } x$$

$$\Rightarrow 1 + 4a \geq 0 \Rightarrow a \geq -\frac{1}{4}$$

$\therefore A, B, D$ is the key

6. Ans (A, C)

For $\Delta A_1 B_1 C_1$ the triangle ABC will be the pedal triangle

$\therefore$ Orthocentre of $\Delta A_1 B_1 C_1$ will be the incentre of ΔABC

$$A_1 = 90^\circ - \frac{A}{2} = \frac{\pi}{3} - \left(\frac{A}{2} - \frac{\pi}{6} \right)$$

$$A_2 = \frac{\pi}{3} - \left(-\frac{A}{2} + \frac{\pi}{12} \right)$$

$$\therefore \lim_{x \rightarrow \infty} A_n = \frac{\pi}{3}$$

$\therefore \Delta A_n B_n C_n$ becomes equilateral

$$B_1 C_1 = R \sin \left(90^\circ - \frac{A}{2} \right) = R \cos \frac{A}{2} = \frac{a}{2 \sin A} \cdot \cos \frac{A}{2}$$

$$= \frac{a}{2 \cdot 2 \sin \frac{A}{2} \cos \frac{A}{2}} \cos \frac{A}{2} = \frac{a}{4} \cos \frac{A}{2}$$

PART-3 : MATHEMATICS

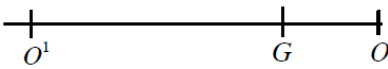
SECTION-I (ii)

7. Ans (A)

$$\vec{SA} + \vec{SB} + \vec{SC} = \vec{OA} + \vec{OB} + \vec{OC} - 3\vec{OS} = 3\vec{OG} - 3\vec{OS} = 3(\vec{SG})$$

$A \rightarrow Q$

$$\vec{OA} + \vec{OB} + \vec{OC} = 3\vec{OG}$$

$$3\vec{G} = 2\vec{O} + \vec{O}'$$


$$|\vec{OO}'| = 3|\vec{OG}|$$

$$\vec{O'A} + \vec{O'B} + \vec{O'C} = \vec{OA} + \vec{OB} + \vec{OC} - (\vec{OO'} + \vec{OO'} + \vec{OO'})$$

$$= 3\vec{OG} - 3\vec{OO'} = \vec{OO'} - 3\vec{OO'} = 2\vec{OO'}$$

$\therefore$ Answer is 2.

$$D) |\vec{OO'} - \vec{OA} + \vec{OB} - \vec{OO'} + \vec{OC} - \vec{OO'}|$$

$$|\vec{OA} + \vec{OB} + \vec{OC} - 2\vec{OA} - \vec{OO'}| = |3\vec{OG} - 2\vec{OA} - \vec{OO'}| = |2\vec{OA}| = 2R = |\vec{AP}|$$

$\therefore$ Answer is 5.

8. Ans (B)

$$(A) |\vec{a} \quad \vec{b} \quad \vec{c}| = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix}$$

$$= (1(-1) - 1(-2) + 1(3))^2 = 16$$

$A \rightarrow r$

$$B) \text{ Equation of OQR is } \begin{vmatrix} x & y & z \\ 1 & 3 & 4 \\ 2 & 1 & -2 \end{vmatrix} = 0$$

$$\Rightarrow x(-10) - y(-10) + z(-5) = 0$$

$$\Rightarrow 2x - 2y + z = 0$$

$$\therefore \text{Distance} = \frac{|16 + 4 - 11|}{3} = \frac{9}{3} = 3$$

C) $(\lambda, \lambda-1, \lambda)$, $(2u-1, u, u)$ be A and B

$\therefore$ Dr's of $\vec{AB}$ are $(2u-\lambda-1, u-\lambda+1, u-\lambda)$ (2, 12)

$$\frac{2u-\lambda-1}{2} = \frac{u-\lambda+1}{1} = \frac{u-\lambda}{2}$$

$$\Rightarrow \lambda = 3$$

$$2u - 2\lambda + 2 = u - \lambda$$

$$u = \lambda - 2 = 1$$

$$A(3, 2, 3) \quad B(1, 1, 1)$$

$$\therefore AB = \sqrt{4 + 1 + 4} = 3$$

D) Let OABC be the tetrahedron

$$\text{Centre of } \vec{OAB} = \frac{\vec{a} + \vec{b}}{3} = G_1. \quad \text{Centre of}$$

$$\vec{OBC} = \frac{\vec{b} + \vec{c}}{3} = G_2$$

$$\therefore G_3 = \frac{\vec{c} + \vec{a}}{3}, \quad G_4 = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$$

$\therefore$ Volume of smaller tetrahedron

$$= \frac{1}{6} \left| \begin{vmatrix} \vec{c} & \vec{a} & \vec{b} \\ 3 & 3 & 3 \end{vmatrix} \right|$$

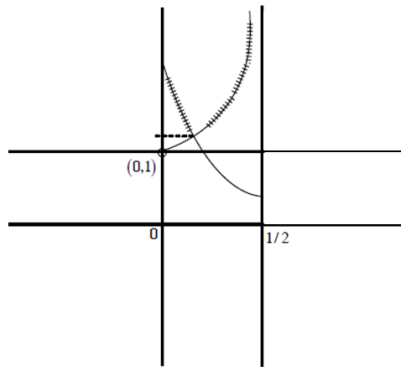
$$\text{Volume of tetrahedron} = \frac{1}{6} |[\vec{a} \quad \vec{b} \quad \vec{c}]|$$

$$\therefore \text{Ratio} = \frac{1}{27}$$

$$\therefore m + n = 28$$

9. Ans (C)

$$0 < x < \frac{1}{2} \Rightarrow 0 < \pi x < \frac{\pi}{2} \Rightarrow 0 < \sin \pi x < 1$$



They intersect at $x = \frac{1}{4}$

$$\therefore \text{Range} = (3 - \sqrt{2}, 2] - \{1\}$$

Numbers integers = 1

Clearly no. of extreme = 1

$$\text{At } x = \frac{1}{4}$$

$f(x)$ is an integer

10. Ans (C)

A) $\sqrt{\sin x}$ is increasing on $\left[0, \frac{\pi}{3}\right]$ so it is one-one and range = $\left[0, \sqrt{\frac{\sqrt{3}}{2}}\right]$

$\therefore$ into

$$\text{B) } f(x_1) = f(x_2) \Rightarrow \frac{x_1 + 3}{x_1 - 1} = \frac{x_2 + 3}{x_2 - 1}$$

$$\Rightarrow -x_1 + 3x_2 - 3 = 3x_1 - 3 - x_2$$

$$\Rightarrow 3(x_1 - x_2) + (x_1 - x_2) = 0 \Rightarrow x_1 = x_2$$

$$y = \frac{x+3}{x-1} \Rightarrow xy - y = x + 3$$

$$\Rightarrow x = \frac{y+3}{y-1}$$

$$x > 1 \Rightarrow \frac{y+3}{y-1} - 1 > 0$$

$$\Rightarrow \frac{4}{y-1} > 0 \Rightarrow y > 1$$

$$\therefore \text{Range} = (1, \infty)$$

$\therefore$ One-One, onto

$$\text{C) } \sin x \text{ is one-one } \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$\therefore$ It is many-one onto

$$f'(x) = \frac{x(x-4)}{(x-2)^2}$$

$\therefore$ It is $\uparrow$ on $(4, \infty)$ and $\downarrow$ in $(2, 4)$

$\therefore$ It is many-one

$$\text{D) } y = \frac{x^2}{x-2} \Rightarrow x^2 - xy + 2y = 0$$

$$\Rightarrow y^2 - 8y \geq 0$$

$$\Rightarrow \text{but } y > 0$$

$$\therefore y \in [8, \infty)$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.00)

$$\begin{aligned}\arg(z^{1/3}) &= \frac{1}{2} \arg(z^2 + \bar{z}z^{1/3}) \\ \arg\left(\frac{z^{1/3}}{z^2 + \bar{z}z^{1/3}}\right) &= 0 \\ \Rightarrow \frac{z^{2/3}}{z^2 + \bar{z}z^{1/3}} &= \frac{\bar{z}^{2/3}}{\bar{z}^2 + z\bar{z}^{1/3}} \\ \Rightarrow z^{2/3} \cdot \bar{z}^2 + z\bar{z}^{2/3}z^{-1/3} &= z^2\bar{z}^{2/3} + \bar{z}^{5/3}z^{1/3} \\ \Rightarrow z^{2/3} \cdot \bar{z}^{2/3}(\bar{z}^{4/3} - z^{4/3}) + z^{1/3}\bar{z}^{1/3}(z^{4/3} - \bar{z}^{4/3}) &= 0 \\ \Rightarrow (\bar{z}^{4/3} - z^{4/3})(|z|^{2/3} - |\bar{z}|^{4/3}) &= 0 \\ \Rightarrow z = \bar{z} \Rightarrow z \text{ is real not possible} \\ \therefore |z|^{2/3}(|z|^{2/3} - 1) &= 0 \Rightarrow |z| = 1\end{aligned}$$

2. Ans (13.00)

Equation of circle is $x^2 + y^2 = a^2e^2$, Let P = (aecosθ, aesinθ)

$$\text{Area of triangle} = 30 \Rightarrow a^2e^2\sin\theta = 30 \quad \dots(1)$$

$$\text{But P lies on the ellipse} \Rightarrow \sin\theta = \frac{1}{e^2} - 1$$

$$\therefore (1) \Rightarrow \frac{289}{4}e^2 \frac{(1 - e^2)}{e^2} = 30$$

$$\Rightarrow e^2 = \frac{169}{289} \Rightarrow e = \frac{13}{17}$$

$$\therefore S_1S_2 = 2 \times \frac{17}{2} \times \frac{13}{17} = 13$$

3. Ans (7.00)

Given

$$\begin{aligned}\det &= \begin{vmatrix} -1 & b & a+2b+2c \\ -1 & c & b+2a+2c \\ -1 & a & c+2a+2b \end{vmatrix} \quad c_1 \rightarrow c_1 + c_2 \\ &= \begin{vmatrix} -1 & b & a+2b+2c \\ 0 & c-b & a-b \\ 0 & a-c & b-c \end{vmatrix} \\ &= a^2 + b^2 + c^2 - ab - bc - ca \\ &= (-1)^2 - 3(-2) = 7\end{aligned}$$

4. Ans (1.00)

$$a_2 = 2(1 + a_1)$$

$$\Rightarrow 1 + a_1 = \frac{a_2}{2}, 1 + a_2 = \frac{a_3}{3} \dots \dots 1 + a_{n-1} = \frac{a_n}{n}$$

Given limit =

$$\begin{aligned}\lim_{n \rightarrow \infty} \left(\frac{1+a_1}{a_1} \right) \left(\frac{1+a_2}{a_2} \right) \dots \dots \left(\frac{1+a_n}{a_n} \right) \\ = \lim_{n \rightarrow \infty} \frac{1}{2} \cdot \frac{1}{3} \dots \dots \frac{1}{n} (1+a_n) = \lim_{n \rightarrow \infty} \frac{(1+a_n)}{n!}\end{aligned}$$

$$\text{But } \frac{a_n}{n!} = \frac{1}{(n-1)!} + \frac{a_{n-1}}{(n-1)!}$$

$$\Rightarrow \frac{a_n}{n!} - \frac{a_{n-1}}{(n-1)!} = \frac{1}{(n-1)!}$$

$$\text{Put } n = 2, 3, 4 \dots \dots \frac{a_2}{2!} - \frac{a_1}{1!} = \frac{1}{1!}$$

$$\frac{a_3}{3!} - \frac{a_2}{2!} = \frac{1}{2!}$$

$$\frac{a_n}{n!} - \frac{a_{n-1}}{(n-1)!} = \frac{1}{(n-1)!}$$

$$\frac{a_n}{n!} - \frac{a_1}{1!} = \frac{1}{1!} + \frac{1}{2!} + \dots \dots + \frac{1}{(n-1)!}$$

$$\therefore \frac{a_n}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots \dots + \frac{1}{(n-1)!}$$

$$\therefore \lim_{n \rightarrow \infty} \left(\frac{1+a_n}{n!} \right) = \lim_{n \rightarrow \infty} \frac{1}{n!} + \lim_{n \rightarrow \infty} \frac{a_n}{n!} = e$$

$$\therefore \text{Required value} = e \times \frac{1}{e} = 1$$

5. Ans (8.00)

$$f(x) + f(y) \leq 0$$

$$\Rightarrow x^2 + 6x + 1 + y^2 + 6y + 1 \leq 0$$

$$\Rightarrow (x+3)^2 + (y+3)^2 \leq 16$$

$$f(x) - f(y)$$

$$\Rightarrow x^2 + 6x + 1 - y^2 - 6y - 1 \leq 0$$

$$\Rightarrow (x^2 - y^2) + 6(x - y) \leq 0$$

$$\Rightarrow (x - y)(x + y + 6) \leq 0$$

Required area is the area of semicircle = 8π

$$\therefore k = 8$$

6. Ans (3.00)

$$\cos^2 \frac{\pi}{8} = \frac{\sqrt{2}+1}{2\sqrt{2}}, \text{ the other root is } \frac{\sqrt{2}-1}{2\sqrt{2}}$$

$$\therefore \text{Sum} = 1 = -p \Rightarrow p = -1$$

$$\therefore [\cos 1 + \sin 1] = \left[\sqrt{2} \sin \left(\frac{\pi}{4} + 1 \right) \right] = 1$$

$$q^2 = \frac{1}{8} \Rightarrow 1 - 1 + 2 \left| \log_2 \frac{1}{2\sqrt{2}} \right| = 3$$

7. Ans (8.00)

 Let $\vec{a} = (\cos x \cos y, \sin y, \sin x \cos y)$,

 $\vec{b} = (1, 2, 2)$

$$\vec{a} \cdot \vec{b} = \cos x \cos y + 2 \sin y + 2 \sin x \cos y$$

$$(\vec{a} \cdot \vec{b})^2 \leq |\vec{a}|^2 |\vec{b}|^2$$

$$\Rightarrow (\cos x \cos y + 2 \sin y + 2 \sin x \cos y)^2 \leq 1.9$$

$$\Rightarrow \cos x \cos y + 2 \sin y + 2 \sin x \cos y \leq 3$$

$$\therefore \frac{\cos x \cos y}{1} = \frac{\sin y}{2} = \frac{\sin x \cos y}{2}$$

$$\tan y = 2 \cos x, \tan y = \sin x$$

$$\tan x = 2, \tan y = \frac{2}{\sqrt{5}}$$

$$\therefore \tan^2 x + 5 \tan^2 y = 4 + 4 = 8$$

8. Ans (4.00)

$$\lim_{n \rightarrow \infty} \left(\frac{3!}{2n!n!} \cdot \frac{n!n!}{(2n)!} \right)^{\frac{1}{n}} = \frac{a}{b}$$

$$y = \lim_{n \rightarrow \infty} \left(\frac{(2n+1)(2n+2) \dots (2n+n)}{(n+1)(n+2) \dots (n+n)} \right)^{\frac{1}{n}} = \frac{a}{b}$$

$$\lim_{n \rightarrow \infty} \left(\frac{\left(2 + \frac{1}{n}\right) \left(2 + \frac{2}{n}\right) \dots \left(2 + \frac{n}{n}\right)}{\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \dots \left(1 + \frac{n}{n}\right)} \right)^{\frac{1}{n}} = \frac{a}{b}$$

$$\log y = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left[\log \left(2 + \frac{r}{n} \right) - \log \left[1 + \frac{r}{n} \right] \right] = \frac{a}{b}$$

$$= \int_0^1 \log \left(\frac{2+x}{1+x} \right) dx = \int_2^3 \log t dt - \int_1^2 \log z dz$$

$$= (t(\log t - 1))_2^3 - (z(\log z - 1))_1^2$$

$$= 3(\log 3 - 1) - 2(\log 2 - 1) - 2(\log 2 - 1) + 1(0 - 1)$$

$$= 3 \log 3 - 4 \log 2 = \log \frac{27}{16}$$

$$\therefore \frac{a}{b} = \frac{27}{16}$$

$$\frac{a+b+1}{a-b} = \frac{27+16+1}{27-16} = \frac{44}{11} = 4$$